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1 Project 4 - Long Memory, ARMA, FIGARCH, and Value-at-Risk

Course: Financial Econometrics

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1.1 Objective

This project studies daily returns of the financial institutions index between 2008 and 2023. It examines long memory, selects an ARMA mean equation, tests residual diagnostics and normality, estimates a FI-GARCH-type volatility model, and computes Value-at-Risk using several methods.

1.2 Libraries and data

The cleaned Excel files contain daily dates and prices. The financial institutions index is stored in `sarmaye.xlsx`; the overall market index is stored in `index.xlsx` and is later used as an external regressor.

```
library(pdR)
library(tidyverse)
library(ggplot2)
library(forecast)
library(tseries)
library(rio)
library(readxl)
library(zoo)
library(lmtest)
library(qqplotr)
library(devtools)
library(fracdiff)
library(MASS)
library(xts)
library(PerformanceAnalytics)
```

```
library(tidyr)
library(rugarch)
library(moments)
library(dplyr)
library(nortest)
library(fBasics)
library(TTR)
library(quantmod)
library(stargazer)
library(ggthemes)
library(gridExtra)

sarmaye <- read_excel("data/sarmaye.xlsx")
ind <- read_excel("data/index.xlsx")
```

1.3 Price and return series

The price series is converted to a daily time series with frequency 235. Returns are defined as first differences of log prices. In the plot, daily returns remain within approximately -10 percent and +10 percent.

```
x <- ts(sarmaye$price, start=c(2008, 12, 06), frequency=235)
ts.plot(x, main="daily price of investment index", ylab="price")

returns <- diff(log(x), lag=1)
returns2 <- ts(returns, start=c(2008, 12, 06), frequency=235)
ts.plot(returns2, main="daily return of investment index",
        ylab="return", ylim=c(-0.1, 0.1))
```

1.4 Long memory

The GPH fractional-d estimate is about $d = 0.234$. Since this value is between 0 and 0.5, the return series is stationary, mean-reverting, and has long memory with finite variance.

```
fdGPH(returns, bandw.exp=0.5)
```

1.5 ARMA order selection

The mean equation is selected by comparing AIC, BIC, and `auto.arima`. The AIC search chooses ARMA(5,5), BIC chooses ARMA(1,2), and `auto.arima` chooses ARIMA(4,0,1). Because the ARMA(5,5) has the highest log-likelihood among the three reported alternatives, it is used for the mean equation.

Method	p	q	Log-likelihood
AIC grid search	5	5	10851.72
BIC grid search	1	2	10835.95
auto.arima	4	1	10841.42

```

AIC <- matrix(nrow=6, ncol=6)
for (i in 1:6) {
  for (j in 1:6) {
    AIC[i,j] <- AIC(arima(returns, c(i-1,0,j-1), method=c("CSS-ML"),
                        optim.control=list(maxit=5000), kappa=1e4), k=2)
  }
}
for (i in 1:6) {
  for (j in 1:6) {
    if (AIC[i,j] == min(AIC)) { p <- i-1; q <- j-1 }
  }
}
arma_sarmaye <- arima(returns, c(p,0,q), method=c("CSS-ML"),
                    optim.control=list(maxit=5000), kappa=1e4)
arma_sarmaye

BIC <- matrix(nrow=6, ncol=6)
for (i in 1:6) {
  for (j in 1:6) {
    BIC[i,j] <- BIC(arima(returns, c(i-1,0,j-1), method=c("CSS-ML"),
                        optim.control=list(maxit=5000), kappa=1e4))
  }
}
for (i in 1:6) {
  for (j in 1:6) {
    if (BIC[i,j] == min(BIC)) { p <- i-1; q <- j-1 }
  }
}
arma_sarmaye2 <- arima(returns, c(p,0,q), method=c("CSS-ML"),
                    optim.control=list(maxit=5000), kappa=1e4)
arma_sarmaye2

auto.arima(returns)

mean.equ <- arma_sarmaye
summary(mean.equ)

```

1.6 Mean-model diagnostics

The ACF and PACF of the ARMA residuals are close to zero. The Box-Pierce and Ljung-Box p-values are both approximately 0.958, so the null hypothesis that residuals are white noise is not rejected at the 5 percent level. This supports the adequacy of the mean equation.

```

mean.residuals <- mean.equ$residuals
acf(mean.residuals, lag.max=8)
pacf(mean.residuals, lag.max=8)

Box.test(mean.residuals, lag=log(3393), c("Box-Pierce"))

```

```
Box.test(mean.residuals, lag=log(3393), c("Ljung-Box"))
```

1.7 Distributional analysis and ARCH effects

The return distribution is right-skewed and leptokurtic. The reported skewness is approximately 0.730 and excess kurtosis is approximately 3.894. The Anderson-Darling normality test gives a p-value below 0.05, so normality is rejected. Tests on squared residuals indicate remaining volatility dependence, which motivates GARCH-type volatility modeling.

```
skewness(returns)
kurtosis(returns)
hist(returns)
ad.test(returns)
plot(density(returns))
chart.QQPlot(returns, distribution="norm")

acf(mean.residuals^2, lag.max=30)
pacf(mean.residuals^2, lag.max=30)
Box.test(mean.residuals^2, lag=log(3393), c("Box-Pierce"))
Box.test(mean.residuals^2, lag=log(3393), c("Ljung-Box"))
```

1.8 FI-GARCH modeling

A FI-GARCH specification is fitted with an ARMA(5,5) mean equation, ARFIMA terms, and skewed Student-t innovations. A second model also includes the overall market index return as an external regressor in the mean equation.

```
garch <- ugarchspec(variance.model=list(model="fiGARCH", garchOrder=c(1,1)),
                    mean.model=list(armaOrder=c(5,5), include.mean=FALSE,
                                     archm=FALSE, archpow=1, arfima=TRUE,
                                     archex=FALSE),
                    distribution.model="sstd")
fit <- ugarchfit(garch, data=returns)
fit
plot(fit, which="all")

z1 <- ts(ind$index)
z2 <- diff(log(z1), lag=1)
garch2 <- ugarchspec(variance.model=list(model="fiGARCH", garchOrder=c(1,1)),
                     mean.model=list(armaOrder=c(5,5), include.mean=FALSE,
                                       archm=FALSE, archpow=1, arfima=TRUE,
                                       archex=TRUE,
                                       external.regressors=matrix(z2)),
                     distribution.model="sstd")
fit2 <- ugarchfit(garch2, data=returns)
fit2
plot(fit2, which="all")
```

1.9 Value-at-Risk

The fitted volatility model is used to plot conditional risk bands. The one-period 95 percent VaR is also computed by modified, Gaussian, and historical methods.

Method	VaR
Modified	-0.01336006
Gaussian	-0.01659899
Historical	-0.01464882

Using the modified VaR as the benchmark, there is a 5 percent probability of losing at least about 1.3 percent in one day when holding the financial institutions index.

```
n <- length(returns)
model.fit <- ugarchfit(spec=garch, data=returns, solver="solnp")
model.fit

qplot(y=returns, x=1:n, geom="point") +
  geom_point(size=0.1) +
  geom_line(aes(y=model.fit@fit$sigma*qdist(distribution="sstd",
                                           shape=3.4125, p=0.05), x=1:n),
            colour="red", size=1.0) +
  geom_line(aes(y=model.fit@fit$sigma*qdist(distribution="sstd",
                                           shape=3.4125, p=0.95), x=1:n),
            colour="red", size=1.0) +
  labs(x=" ", y="Daily Return", title="Value at Risk Comparison")

sarmaye <- data.frame(cbind(year=substr(sarmaye[,1],1,4),
                           month=substr(sarmaye[,1],5,6),
                           day=substr(sarmaye[,1],7,8), sarmaye))
sarmaye$date <- as.Date(sarmaye$date)
price_xts <- xts(sarmaye$price, order.by=sarmaye$date)
return_xts <- diff(log(price_xts), lag=1)

mean(returns)
sd(returns)
skewness(returns)
kurtosis(returns)

VaR(R=return_xts, p=0.95, method=c("modified"), invert=TRUE,
    mu=0.001377912, sigma=0.01093079, m3=0.7302357, m4=3.893968)
VaR(R=return_xts, p=0.95, method=c("gaussian"), invert=TRUE,
    mu=0.001377912, sigma=0.01093079, m3=0.7302357, m4=3.893968)
VaR(R=return_xts, p=0.95, method=c("historical"), invert=TRUE,
    mu=0.001377912, sigma=0.01093079, m3=0.7302357, m4=3.893968)
```

1.10 Conclusion

The return series is stationary but long-memory. The ARMA(5,5) mean equation is adequate according to residual autocorrelation diagnostics, but returns are non-normal and volatility clustering remains. FI-GARCH modeling and VaR estimation are therefore appropriate tools for the risk analysis.